

Lecture No. 42-45 : Digital Electronics

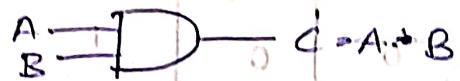
Gates:

① OR Gate:



A	B	C
1	1	1
1	0	1
0	1	1
0	0	0

② AND Gate:



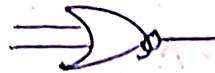
A	B	C
0	0	0
0	1	0
1	0	0
1	1	1

③ NOT Gate: (Single output gate)



A	$\bar{A}$
0	1
1	0

④ NOR Gate:



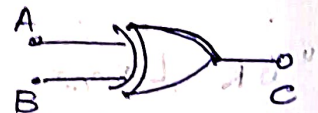
A	B	C
0	0	1
1	0	0
0	1	0
1	1	0

⑤ NAND Gate:



A	B	C
0	0	1
1	0	1
0	1	1
1	1	0

⑥ XOR Gate:



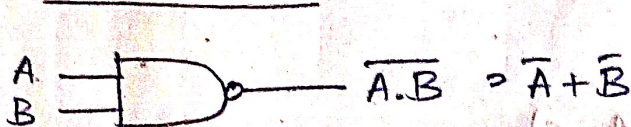
A	B	C
0	0	0
1	0	1
0	1	1
1	1	0

← last digit

2 = (1+1) → in binary → 10

$$1 \times 2^1 + 0 \times 2^0$$

* NAND Gate:



Equivalent Circuits

A	B	$\overline{A \cdot B}$	$\overline{A} \cdot \overline{B}$
1	1	0	0
1	0	1	1
0	1	1	1
0	0	1	1

→ verify using truth table

Similarly: $\overline{A+B} = \overline{A} \cdot \overline{B}$

Laws of Boolean Algebra

(i) $0 = 1$; (ii) $1 = 0$

(iii) If $A = 0$, $\overline{A} = 1$; (iv) If $A = 1$ & $\overline{A} = 0$

(v) $\overline{\overline{A}} = A$

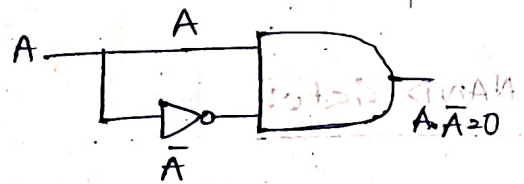
Complementing Law

"NOT"

"AND" Laws

(vi) $A \cdot 0 = 0$; (vii) $A \cdot 1 = A$

(viii) $A \cdot A = A$; (ix) $A \cdot \overline{A} = 0$ (always)



"OR" Laws

(x) $A + 0 = A$; (xi) $A + 1 = 1$

(xii) $A + A = A$; (xiii) $A + \overline{A} = 1$ (always)

Commutative Laws

(xiv) $A + B = B + A$; (xv) $A \cdot B = B \cdot A$

Associative Laws

(xvi) $A + (B + C) = (A + B) + C$; (xvii) $A \cdot (B \cdot C) = (A \cdot B) \cdot C$

Distributive Laws

(xviii) $A \cdot (B + C) = (A \cdot B) + (A \cdot C)$

(xix) $A + (B \cdot C) = (A + B) \cdot (A + C)$

(xx) $A + (A \cdot B) = A + B$ → other forms $\overline{A} + (A \cdot B) = \overline{A} + B$
 $\overline{B} + (A \cdot B) = \overline{B} + A$

* DeMorgan's Theorem (General form)

$$(xxi) \overline{A+B+C+\dots} = \bar{A} \cdot \bar{B} \cdot \bar{C} \cdot \dots$$

$$(xxii) \overline{A \cdot B \cdot C \cdot D \cdot \dots} = \bar{A} + \bar{B} + \bar{C} + \dots$$

* Procedure to Apply DeMorgan's Theorem

- (1) Complement all variables including output variables
- (2) Change all AND to OR gates
- (3) Change all OR to AND gates
- (4) Complement both sides of equations to obtain the original resultant variables.

☹

* Any arbitrary logic form can be expressed as:

(i) SUM OF PRODUCTS FORM (SOP)

(ii) PRODUCT OF SUMS FORM (POS)

Lets consider the funcⁿ:

$$F = (A+BC)(B+\bar{C}A) \quad \text{--- (1)}$$

(i) Sum of product form (SOP)

$$F = (A+BC)(B+\bar{C}A)$$

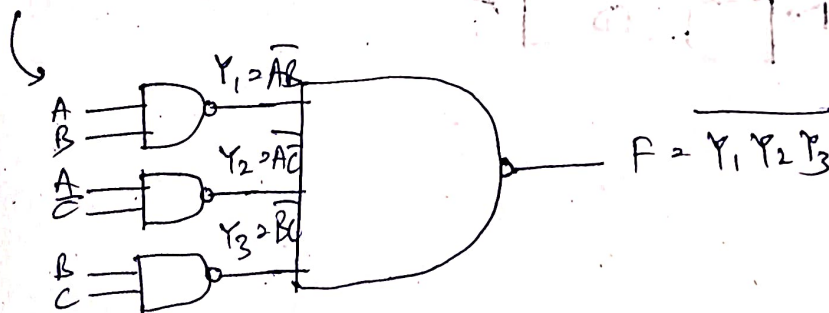
$$= A(B+\bar{C}A) + BC(B+\bar{C}A) \rightarrow 0$$

$$= AB + A\bar{C}A + BCB + BC\bar{C}A$$

$$\begin{bmatrix} A \cdot A = A \\ B \cdot B = B \end{bmatrix}$$

$$= AB + A\bar{C} + BC + 0$$

$$= AB + A\bar{C} + BC \quad \text{--- (2)}$$



Qii) POS ;

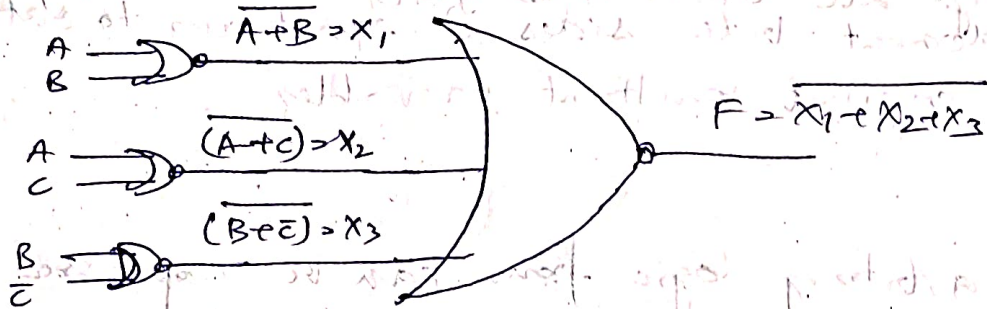
$$F = (A + \bar{B}C) (B + \bar{C}A)$$

$$= (A + \bar{B})(A + C) (B + \bar{C})(B + A)$$

$$= \underbrace{(A + \bar{B})(A + C)}_{= (A + \bar{B}C)} (B + \bar{C})(B + A)$$

$$= (A + \bar{B}C) (B + \bar{C})(B + A)$$

$$= (A + \bar{B}C) (A + C) (B + \bar{C})$$



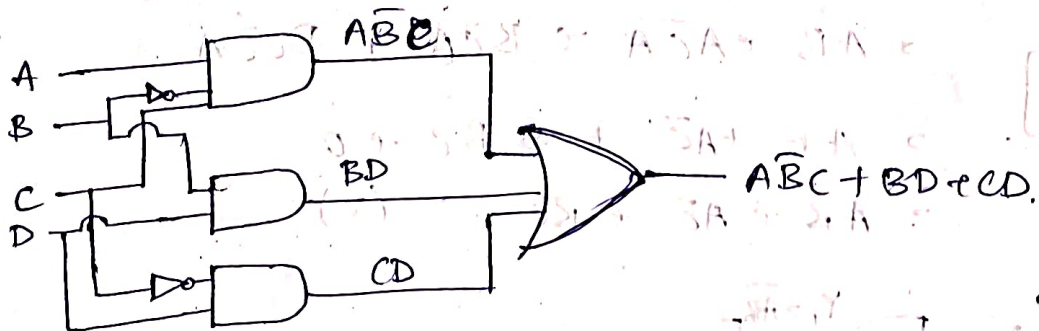
Lecture 44 : 20/11/25

• DeMorgan's Theorem : breaking of NAND into OR.

$$\bar{A}\bar{B}\bar{C}D + A\bar{B}\bar{C}D + \bar{C}\bar{D} + A\bar{D}$$

$$= (\bar{A} + B) \bar{C}\bar{D} + A\bar{B}\bar{C}D + \bar{C}\bar{D} + A\bar{D}$$

$$= \bar{A}\bar{C}\bar{D} + \bar{B}\bar{C}\bar{D} + A\bar{B}\bar{C}D + \bar{C}\bar{D} + A\bar{D} \rightarrow \text{SOP format}$$



Convert an SOP to Truth Table

Clues: Look for ones that have "zero" as output

Look for ones that have "one" as output

A	B	C	$A \cdot B \cdot C$	$\bar{A} \bar{B} C$	$A \bar{B} \bar{C}$	$AB \bar{C}$	$ABC + \bar{A} \bar{B} C + A \bar{B} \bar{C} + AB \bar{C}$
0	0	0	0	0	0	0	0
0	0	1	0	0	0	0	0
0	1	0	0	0	1	0	1
0	1	1	0	0	0	0	0
1	0	1	0	1	0	0	1
1	1	0	0	0	1	0	1
1	1	1	1	0	0	0	1
1	0	0	0	0	0	0	0

A	B	C	X
0	0	0	0
0	0	1	1 $\rightarrow \bar{A} \cdot \bar{B} \cdot C$
0	1	0	0
0	1	1	1 $\rightarrow \bar{A} B C$
1	0	0	1 $\rightarrow A \bar{B} \bar{C}$
1	0	1	0
1	1	0	1 $\rightarrow A B \bar{C}$
1	1	1	0

$\rightarrow$ "OR" gate
Sum gate

POS (Product of Sums)

$$(\bar{A} + B + C + \bar{D})(A + \bar{B} + D)(C + \bar{D})(A + \bar{D})$$

POS into truth table $\rightarrow$ check: look for 2 zeros!

$$(A \oplus B \oplus C) (A \oplus \bar{B} \oplus C) (\bar{A} \oplus B \oplus \bar{C})$$

A	B	C	X
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

$$A \oplus B \oplus C$$

$$A \oplus \bar{B} \oplus C$$

$$\bar{A} \oplus B \oplus \bar{C}$$

$$\bar{A} \oplus \bar{B} \oplus \bar{C}$$

$$(A \oplus B \oplus C) (A \oplus \bar{B} \oplus C) (\bar{A} \oplus B \oplus \bar{C}) (\bar{A} \oplus \bar{B} \oplus \bar{C})$$

Lecture 45: 21/11/15.

Karnaugh Map!

Truth Table: (2 Bits) (A)

A	B	X
0	0	$S_0 = 0$
0	1	$S_1 = 0$
1	0	$S_2 = 0$
1	1	$S_3 = 1$

A \ B	0	1
0	$S_0 = 0$	$S_1 = 0$
1	$S_2 = 0$	$S_3 = 1$

(B) 3 Bits:

Eg 1

A	B	C	X
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

A \ B \ C	00	01	11	10
0	0	1	1	1
1	1	0	0	1

C' changes
but AB doesn't
change AB

$$A=0, B=1, C=0; 1 \cdot 1 + 0 \cdot 0 = 1$$

$$A=1, B=1, C=1; 1 \cdot 0 + 1 \cdot 0 = 0$$

$$A=1, B=0, C=1; 0 \cdot 0 + 1 \cdot 1 = 1$$

$$F = BC + AB$$

$$A \rightarrow 1, B \rightarrow 1, C \rightarrow 0$$

$$\rightarrow ABC = 1$$

Eg 2

A	B	C	X
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

(i)
 $\downarrow$
 $\bar{A}C$

A \ B \ C	0	1
00	0	1
01	1	1
11	1	0
10	0	1

(ii)
 $\downarrow$
 $\bar{A}B$

$$F = \bar{A}B + \bar{A}C + ABC$$

Eg 3:
ABCD

CD \ AB	00	01	11	10
00	0	0	1	0
01	0	0	1	1
11	0	0	0	1
10	1	1	0	0

$\rightarrow ABC + A\bar{B}D + \bar{A}CD$

Eg 4:

CD \ AB	00	01	11	10
00	1	0	0	0
01	1	0	1	1
11	1	0	1	1
10	1	0	0	0

c changes
but x same

$\bar{A}\bar{B}$ as

x same

though C & D changes!

B changes
but x same

$\textcircled{AD}$

$F = \bar{A}\bar{B} + AD$